

Minorizing measures and truncations of measurable lattice-path matroids

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Abstract

We study the minorizing-measure sets of the measurable lattice-path matroids recently introduced by Bérczi, Imolay, and Schweitzer. Such a matroid is specified by a closed set of times and by lower and upper cumulative paths. We first prove that its set of basic minorizing measures is exactly the weak-star compact corridor of fractional path densities. The proof is constructive: after moving a fractional path slightly toward a genuine basis path, we purify it on the components of the strict interior by a countable endpoint-preserving bang–bang construction.

We then determine the full minorizing-measure set. A density belongs to it if and only if all interval increments satisfy the inequalities forced by the two boundary paths. The nontrivial direction is proved by an explicit residual-capacity time change that extends the density to a basic minorizing measure. As a consequence, the basic minorizing measures of every real truncation are described by the same interval inequalities together with one total-mass equation. We also record the contact-set criterion for extreme points, show directly that every extreme point is weak-star exposed, and give a bounded laminar-composition application.

The finite-dimensional shadows of the corridor and interval descriptions are classical lattice-path and positroid polytope phenomena. The contribution here is their exact atomless measurable counterpart, including arbitrary closed time sets and real truncation levels.

1 Introduction

Measurable matroids provide a measure-theoretic analogue of finite matroids on standard atomless measure spaces. Bérczi, Imolay, and Schweitzer established equivalent independent-set, basis, rank, and closure descriptions, introduced measurable versions of standard matroid classes and operations, and proved that independent-set and basis measures are weak-star dense in the corresponding minorizing-measure sets [1]. Their examples include measurable lattice-path matroids: a basis is represented by a zero–one derivative whose cumulative path stays between two prescribed boundaries.

The purpose of this paper is to make the associated fractional geometry explicit. There are two distinct convex sets:

1. the basic minorizing measures, which have the full matroid rank as total mass; and

2. all minorizing measures, whose total mass is allowed to vary.

For the first set, the expected candidate is the obvious path corridor. In a finite lattice-path matroid the analogous base-polytope description is classical [2, 3]. In the atomless setting, however, it is not enough to observe that every basis lies in the corridor: one must prove that every fractional corridor path is a weak-star limit of genuine basis paths. We give a self-contained purification proof that remains valid for an arbitrary closed set of constrained times, including infinitely many complementary gaps and Cantor-type contact sets.

The full minorizing-measure set is subtler. Prefix lower bounds no longer make sense without fixed total mass. They are replaced by interval inequalities

$$F(t) - F(s) \leq b(t) - a(s) \quad (s \leq t),$$

where F is the cumulative density. We prove that these inequalities are not merely necessary: they are sufficient. The proof builds, from the unused pointwise capacity $1 - f$, a complementary path that raises the density to a point in the basic minorizing-measure set. This yields the following informal form of the main result:

The complete independence relaxation of a measurable lattice-path matroid is its interval relaxation, and every real truncation is obtained by taking a mass slice.

The extreme-point criterion for the fixed-mass corridor is closely related to classical extreme-point results in separated continuous linear programming [4, 5]. We include a short direct proof because it identifies the exact contact set and gives a concrete exposing functional without relying on a claimed theorem-level equivalence. We do not claim a new general bang-bang or continuous-control theorem: related purification mechanisms have a substantial history [6, 7].

The paper is organized as follows. Section 2 gives the setup. Section 3 proves the basic-minorizing-measure identity. Section 4 determines the full minorizing-measure set and all truncation slices. Section 5 treats extreme and exposed points. Section 6 gives a finite-depth laminar-composition application and carefully separates it from the main contribution. Section 7 records the scope and remaining open boundary.

2 Setup

2.1 Minorizing measures

Let (J, Σ, μ, r) be a measurable matroid in the sense of [1]. Throughout, α denotes a finite nonnegative charge on (J, Σ) . Its set of minorizing measures and its set of basic minorizing measures are

$$\text{mm}(r) = \{\alpha : 0 \leq \alpha(X) \leq r(X) \text{ for every } X \in \Sigma\},$$

and

$$\text{bmm}(r) = \{\alpha \in \text{mm}(r) : \alpha(J) = r(J)\}.$$

Since $r \leq \mu$, every such measure is absolutely continuous with respect to μ and has a density in the order interval $0 \leq f \leq 1$. We use the weak-star topology of $L^\infty(\mu) = (L^1(\mu))^*$. Proposition 5.6 and Corollary 5.7 of [1] state that independent-set measures are weak-star dense in $\text{mm}(r)$ and basis measures are weak-star dense in $\text{bmm}(r)$.

2.2 Measurable lattice-path data

Let $K \subseteq [0, 1]$ be closed and contain 0 and 1. Fix $0 \leq R \leq 1$ and let

$$a, b : K \longrightarrow [0, R]$$

be nondecreasing 1-Lipschitz functions satisfying

$$a(0) = b(0) = 0, \quad a(1) = b(1) = R, \quad a(t) \leq b(t).$$

Assume that the basis family

$$\mathcal{B}_K(a, b) = \{B \subseteq [0, 1] : \lambda(B) = R, a(t) \leq \lambda(B \cap [0, t]) \leq b(t) \quad (t \in K)\}$$

is nonempty. By Theorem 3.13 of [1], it is the basis family of a measurable matroid, which we denote by $M_K(a, b)$. Its rank function is denoted by r .

For $f \in L^\infty([0, 1])$, set

$$F_f(t) = \int_0^t f(x) dx.$$

We use two convex relaxations. The fixed-mass path corridor is

$$\mathcal{P}_K(a, b) = \{f : 0 \leq f \leq 1, F_f(1) = R, a(t) \leq F_f(t) \leq b(t) \quad (t \in K)\}.$$

The interval relaxation is

$$\mathcal{I}_K(a, b) = \{f : 0 \leq f \leq 1, F_f(t) - F_f(s) \leq b(t) - a(s) \quad (s, t \in K, s \leq t)\}.$$

Both sets are weak-star compact. Indeed, the order interval $0 \leq f \leq 1$ is weak-star compact, and every cumulative evaluation $f \mapsto F_f(t)$ is weak-star continuous.

3 The basic minorizing-measure corridor

The first main result identifies the fixed-mass relaxation exactly.

Theorem 3.1 (corridor identity). *For every nonempty measurable lattice-path matroid $M_K(a, b)$,*

$$\text{bmm}(M_K(a, b)) = \{f\lambda : f \in \mathcal{P}_K(a, b)\}.$$

The inclusion from left to right follows immediately from the weak-star density of basis measures. The reverse inclusion is the content of the following purification argument.

Proof.

3.1 Completing the obstacles across the gaps

Let (s, t) be a component of $[0, 1] \setminus K$. Define on this gap

$$\ell(x) = \max\{a(s), a(t) - (t - x)\},$$

and

$$u(x) = \min\{b(s) + x - s, b(t)\}.$$

On K , put $\ell = a$ and $u = b$. These definitions give continuous, nondecreasing, 1-Lipschitz functions on $[0, 1]$. Moreover, a nondecreasing 1-Lipschitz path F satisfies the constraints at the endpoints of a gap if and only if

$$\ell(x) \leq F(x) \leq u(x)$$

throughout the gap. Thus $\mathcal{P}_K(a, b)$ is exactly the set of derivatives of absolutely continuous paths F such that

$$F(0) = 0, \quad F(1) = R, \quad 0 \leq F' \leq 1, \quad \ell \leq F \leq u.$$

Because $\mathcal{B}_K(a, b)$ is nonempty, there is a path H_0 in this corridor with $H'_0 \in \{0, 1\}$ almost everywhere.

3.2 A local endpoint-preserving purification

We first record an elementary lemma.

Lemma 3.2. *Let Y be absolutely continuous on a compact interval $[p, q]$ and satisfy $0 \leq Y' \leq 1$ almost everywhere. For every $\eta > 0$, there exists an absolutely continuous W such that*

$$W(p) = Y(p), \quad W(q) = Y(q), \quad W' \in \{0, 1\} \text{ a.e.},$$

and $\|W - Y\|_\infty < \eta$.

Proof. Choose a mesh of width smaller than η . On a mesh interval $[r, s]$, let $\Delta = Y(s) - Y(r)$. Since $0 \leq \Delta \leq s - r$, define W to move with slope 1 for time Δ and with slope 0 for the rest of the interval. The new and old paths agree at all mesh points and both remain between the same two endpoint values on each mesh interval. The uniform error is therefore smaller than the mesh width. $\square$

3.3 Countable purification inside the corridor

Fix $f \in \mathcal{P}_K(a, b)$ and write $F = F_f$. For $0 < \delta < 1$, set

$$F_\delta = (1 - \delta)F + \delta H_0.$$

Let

$$Z = \{x : F_\delta(x) = \ell(x) \text{ or } F_\delta(x) = u(x)\}.$$

This is closed. If, for example, $F_\delta = \ell$ at a point, then both $F - \ell$ and $H_0 - \ell$ are nonnegative there and their strict convex combination is zero. Hence $F = H_0 = \ell$ at that point. The analogous statement holds on the upper contact set. The zero-set derivative lemma for absolutely continuous functions now implies

$$F'_\delta = H'_0 \in \{0, 1\} \quad \text{almost everywhere on } Z.$$

The complement of Z is a countable union of disjoint open intervals. In each such interval choose a bi-infinite sequence of subdivision points tending to both endpoints. Every compact cell I_j between consecutive subdivision points lies strictly inside the corridor, so

$$\rho_j = \min_{I_j} \{F_\delta - \ell, u - F_\delta\} > 0.$$

Apply Lemma 3.2 on I_j with mesh width smaller than $\min\{\varepsilon, \rho_j\}$. The replacement agrees with F_δ at the cell endpoints, remains in the corridor, and has zero-one derivative almost everywhere. Keep F_δ on Z and at all subdivision points. This defines a continuous path H on $[0, 1]$.

For completeness, we verify the global regularity rather than merely the cellwise regularity. Let S be the union of Z and all subdivision points. It is closed: new accumulation points of subdivision points lie in Z . If $s < t$ lie in different cells, let $r \in S$ be the right endpoint of the cell containing s , with the convention $r = s$ when $s \in S$, and let $q \in S$ be the left endpoint of the cell containing t , with the convention $q = t$ when $t \in S$. These conventions also remove any ambiguity when an endpoint is adjacent to more than one cell. Since $H = F_\delta$ on S ,

$$0 \leq H(t) - H(s) \leq (t - q) + F_\delta(q) - F_\delta(r) + (r - s) \leq t - s.$$

Thus H is globally nondecreasing and 1-Lipschitz, hence absolutely continuous. On $[0, 1] \setminus S$ its derivative is zero-one by construction. On S , the function $H - F_\delta$ vanishes; its derivative is therefore zero almost everywhere on S . The set $S \setminus Z$ is countable, while F'_δ is zero-one almost everywhere on Z . Consequently

$$H' = \mathbf{1}_B$$

almost everywhere for a measurable set B . The endpoint and corridor conditions show that $B \in \mathcal{B}_K(a, b)$. Moreover,

$$\|H - F\|_\infty \leq \varepsilon + \delta.$$

Letting $\varepsilon, \delta \downarrow 0$ gives basis paths converging uniformly to F .

3.4 Weak-star convergence of the derivatives

Suppose $H_n \rightarrow F$ uniformly and $0 \leq H'_n, F' \leq 1$. For every $\varphi \in C^1([0, 1])$, integration by parts gives

$$\int_0^1 \varphi(H'_n - F') = \varphi(1)(H_n - F)(1) - \varphi(0)(H_n - F)(0) - \int_0^1 \varphi'(H_n - F),$$

which tends to zero. Density of $C^1([0, 1])$ in $L^1([0, 1])$, together with the uniform L^∞ bound on the derivatives, yields $H'_n \xrightarrow{*} F'$. Hence $f\lambda$ lies in the weak-star closure of the basis measures, which is $\text{bmm}(M_K(a, b))$ by [1]. This completes the proof of Theorem 3.1. $\square$

4 The full independence relaxation and all truncations

We now allow the total mass to vary.

Theorem 4.1 (interval description of all minorizing measures). *For every nonempty measurable lattice-path matroid $M_K(a, b)$,*

$$\text{mm}(M_K(a, b)) = \{f\lambda : f \in \mathcal{I}_K(a, b)\}.$$

Proof of necessity. Let I be independent. It extends to a basis B . For $s \leq t$ in K ,

$$\lambda(I \cap (s, t]) \leq \lambda(B \cap (s, t]) = F_B(t) - F_B(s) \leq b(t) - a(s).$$

It follows that the rank of $(s, t]$ is at most $b(t) - a(s)$. Every minorizing measure therefore satisfies the interval inequalities. Its density also satisfies $0 \leq f \leq 1$.

Proof of sufficiency.

Fix $f \in \mathcal{I}_K(a, b)$, set $F = F_f$, and write

$$m = F(1), \quad C(t) = t - F(t) = \int_0^t (1 - f(x)) dx.$$

The function C is nondecreasing and 1-Lipschitz. Put $U = 1 - m$ and $S = C(K) \subseteq [0, U]$. For $u \in S$, define

$$A(u) = \sup\{a(t) - F(t) : t \in K, C(t) = u\},$$

and

$$B(u) = \inf\{b(t) - F(t) : t \in K, C(t) = u\}.$$

The fibers are compact. We claim that for $u \leq v$ in S ,

$$A(u) \leq B(v), \quad A(v) - B(u) \leq v - u. \tag{4.1}$$

Take $s, t \in K$ with $C(s) = u$ and $C(t) = v$. If $u < v$, monotonicity of C forces $s < t$. The interval inequality gives

$$a(s) - F(s) \leq b(t) - F(t),$$

while monotonicity and the 1-Lipschitz property of a give

$$a(t) - b(s) \leq t - s.$$

After subtracting $F(t) - F(s)$, the latter becomes

$$a(t) - F(t) - (b(s) - F(s)) \leq C(t) - C(s) = v - u.$$

Taking the appropriate supremum and infimum proves (4.1). If $u = v$, points in the same fiber may occur in either order. Applying the first calculation in the forward order and the 1-Lipschitz calculation in the reverse order gives both inequalities with right-hand side zero. Thus (4.1) holds for non-singleton fibers as well.

Define, for $0 \leq u \leq U$,

$$y(u) = \sup_{v \in S} \{A(v) - \max(v - u, 0)\}. \quad (4.2)$$

Each function in the supremum is nondecreasing and 1-Lipschitz, so the same is true of y . Choosing $v = u$ gives $A(u) \leq y(u)$ for $u \in S$. If $v \leq u$, the first inequality in (4.1) gives $A(v) \leq B(u)$; if $v \geq u$, the second gives $A(v) - (v - u) \leq B(u)$. Hence

$$A(u) \leq y(u) \leq B(u) \quad (u \in S). \quad (4.3)$$

The endpoint fibers give

$$A(0) = B(0) = 0, \text{ and } A(U) = B(U) = R - m.$$

For the first pair, $C(t) = 0$ implies $F(t) = t$, and the interval inequality from 0 to t , together with $b(t) \leq t$, forces $b(t) = t$. For the second pair, $C(t) = U$ means that C is constant from t to 1; the interval inequality and the Lipschitz boundary conditions then squeeze both residual boundaries to $R - m$, with equality at $t = 1$. Equations (4.1)–(4.2) consequently give

$$y(0) = 0, \text{ and } y(U) = R - m. \quad (4.4)$$

Set

$$H(t) = y(C(t)).$$

Both y and $u \mapsto u - y(u)$ are nondecreasing. Therefore H and $C - H$ are nondecreasing absolutely continuous functions. It follows that

$$0 \leq H' \leq C' = 1 - f$$

almost everywhere. For $t \in K$, (4.3) gives

$$a(t) - F(t) \leq H(t) \leq b(t) - F(t).$$

Thus

$$g = f + H'$$

satisfies $0 \leq g \leq 1$, has total mass R by (4.4), and its cumulative function $F + H$ belongs to the corridor $\mathcal{P}_K(a, b)$. Theorem 3.1 gives $g\lambda \in \text{bmm}(M_K(a, b))$.

Finally, $0 \leq f \leq g$. Hence, for every measurable X ,

$$\int_X f d\lambda \leq \int_X g d\lambda \leq r(X).$$

Therefore $f\lambda \in \text{mm}(M_K(a, b))$. □

Corollary 4.2 (all real truncations). *Let $0 \leq c \leq R$, and let $T_c M_K(a, b)$ denote the c -truncation. Then*

$$\text{bmm}(T_c M_K(a, b)) = \left\{ f\lambda : f \in \mathcal{I}_K(a, b), \int_0^1 f = c \right\}.$$

Proof. The rank of a truncation is $r_c(X) = \min\{r(X), c\}$ [1]. Consequently,

$$\text{mm}(T_c M) = \{\alpha \in \text{mm}(M) : \alpha(J) \leq c\}.$$

Indeed, the forward inclusion follows from $r_c \leq r$ and $r_c(J) = c$; conversely, $\alpha(X) \leq r(X)$ and $\alpha(X) \leq \alpha(J) \leq c$ imply $\alpha(X) \leq r_c(X)$. Taking the mass- c slice and applying Theorem 4.1 gives the claim. The endpoint cases $c = 0$ and $c = R$ are included. □

When $c = R$, the interval inequalities with $(s, t) = (0, t)$ and $(t, 1)$ recover the two prefix inequalities, so Corollary 4.2 specializes to Theorem 3.1.

5 Extreme and exposed points

For $f \in \mathcal{P}_K(a, b)$, define

$$T_a(f) = \{t \in K : F_f(t) = a(t)\},$$

$$T_b(f) = \{t \in K : F_f(t) = b(t)\},$$

and $T(f) = T_a(f) \cup T_b(f)$. These sets are closed.

Theorem 5.1 (contact-set criterion). *For $f \in \mathcal{P}_K(a, b)$, the following are equivalent:*

1. $f\lambda$ is an extreme point of $\text{bmm}(M_K(a, b))$;
2. $f\lambda$ is a weak-star exposed point of $\text{bmm}(M_K(a, b))$;

3. f belongs to $\{0, 1\}$ almost everywhere on $[0, 1] \setminus T(f)$.

Proof of necessity. By Theorem 3.1 it suffices to work in $\mathcal{P}_K(a, b)$. If the fractional part of f outside $T(f)$ has positive measure, then some compact interval contained in one component of $[0, 1] \setminus T(f)$ contains a positive-measure set on which $\eta \leq f \leq 1 - \eta$ for some $\eta > 0$. On the constrained times in that compact interval, both relevant slacks have a positive uniform lower bound. Choose disjoint equal-measure sets E_+ and E_- in the fractional region and put $h = \mathbf{1}_{E_+} - \mathbf{1}_{E_-}$. Its cumulative integral vanishes outside the chosen interval. For sufficiently small $\varepsilon > 0$, both $f + \varepsilon h$ and $f - \varepsilon h$ remain in the corridor. Hence f is not extreme.

Proof of sufficiency.

Suppose $f = (g_1 + g_2)/2$ with $g_i \in \mathcal{P}_K(a, b)$. Put $d = (g_1 - g_2)/2$ and $D(t) = \int_0^t d$. The box constraints force $d = 0$ almost everywhere where $f = 0$ or $f = 1$, hence almost everywhere off $T(f)$. At a point of $T_a(f)$ or $T_b(f)$, equality in the averaged active constraint forces both g_i to take the same cumulative value as f . Therefore $D = 0$ on $T(f)$. An absolutely continuous function has derivative zero almost everywhere on its zero set, so $d = 0$ almost everywhere on $T(f)$ as well. Thus $g_1 = g_2 = f$.

Exposedness.

Choose Borel probability measures ν_a and ν_b with full support on the two nonempty compact contact sets. Define

$$q_b(x) = \nu_b(T_b(f) \cap [x, 1]), \quad q_a(x) = \nu_a(T_a(f) \cap [x, 1]).$$

On the complement of $T(f)$, choose a zero-one representative of f and set $c_f = 1$ where $f = 1$ and $c_f = -1$ where $f = 0$; put $c_f = 0$ on $T(f)$. Then

$$w = c_f + q_b - q_a \in L^1([0, 1]).$$

For every $g \in \mathcal{P}_K(a, b)$, the box constraints and Fubini's theorem give

$$\int c_f g \leq \int c_f f,$$

$$\int q_b g = \int_{T_b(f)} F_g(t) d\nu_b(t) \leq \int_{T_b(f)} b(t) d\nu_b(t) = \int q_b f,$$

and similarly $-\int q_a g \leq -\int q_a f$. If equality holds in their sum, every nonnegative gap is zero. The first equality fixes $g = f$ off $T(f)$. Full support and continuity force all cumulative contact equalities, and the zero-set derivative lemma then fixes $g = f$ on $T(f)$ almost everywhere. Thus w uniquely exposes f . If an explicit nonzero exposing representative is desired in a degenerate case, add the constant function 1; the total mass is fixed on the corridor. $\square$

The extreme/basic part of this theorem is analogous to classical SCLP extreme-point criteria [4, 5]. The direct argument above is self-contained; the references provide background rather than a claimed exact theorem-level reduction.

Corollary 5.2 (integrality for a fixed corridor). *Let $\dot{a}_K$ and $\dot{b}_K$ denote the intrinsic almost-everywhere derivatives on K : take any Lipschitz extensions to $[0, 1]$; their derivatives agree*

almost everywhere on K . Every extreme point of $\text{bmm}(M_K(a, b))$ is induced by a basis if and only if

$$\dot{a}_K, \dot{b}_K \in \{0, 1\} \quad \text{almost everywhere on } K.$$

Proof. Theorem 5.1 already forces zero–one values off the contact set. On the lower and upper contact sets, the zero-set derivative lemma gives $f = \dot{a}_K$ and $f = \dot{b}_K$, respectively, almost everywhere. This proves sufficiency.

Conversely, if $\dot{b}_K$ is fractional on a positive-measure subset, extend b over each complementary gap by moving first with slope 1 and then with slope 0, while keeping $F = b$ on K . The resulting path belongs to the corridor, is zero–one off K , and is in upper contact throughout K . Theorem 5.1 makes its derivative an exposed point, but it is fractional on a positive-measure subset of K . The lower boundary is handled analogously. $\square$

6 A bounded laminar-composition application

This section records a useful closure consequence but is not a separate main contribution.

Let T be a finite rooted tree whose internal nodes have finite arity. Its leaves carry pairwise disjoint atomless standard probability spaces, each identified measure preservingly with $[0, 1]$. At a leaf ℓ , let $K_\ell \subseteq [0, 1]$ be closed and contain 0, 1, and let $b_\ell : K_\ell \rightarrow [0, \infty)$ be an effective, nondecreasing, 1-Lipschitz upper capacity with $b_\ell(0) = 0$. If one starts from an arbitrary capacity presentation, it must first be replaced by its effective rank representative: for a nested family $\mathcal{D}$ with raw capacity b , use the rank-effective capacity

$$\beta(C) = \inf_{D \in \mathcal{D}} \{b(D) + \mu(C \setminus D)\} \quad (C \in \mathcal{D}),$$

and then parameterize the resulting closed chain by K_ℓ . The contact-set statement below is not asserted for a raw discontinuous or redundant presentation.

Let

$$C_\ell = \left\{ f_\ell : 0 \leq f_\ell \leq 1, \int_0^t f_\ell \leq b_\ell(t) \quad (t \in K_\ell) \right\}.$$

Every internal node is labelled either by the direct sum of its children or by a truncation of the matroid represented by its unique child; every truncation capacity c_v is a nonnegative real number. Consecutive direct sums may be merged and consecutive truncations replaced by the smaller capacity. If $L(v)$ is the set of leaves below a truncation node v , define

$$Q_T = \left\{ (f_\ell) : f_\ell \in C_\ell, \sum_{\ell \in L(v)} \int f_\ell \leq c_v \text{ for every truncation node } v \right\}.$$

Proposition 6.1. *For the measurable matroid M_T obtained from this tree,*

$$\text{mm}(M_T) = Q_T,$$

and

$$\text{bmm}(M_T) = \left\{ f \in Q_T : \sum_{\ell} \int f_{\ell} = r_T(J) \right\}.$$

Proof. At a leaf, the effective nested rank formula says that the minorizing-measure set is exactly C_{ℓ} . Minorizing-measure sets multiply under direct sum. Under a c -truncation they are intersected with the total-mass halfspace of level c , by the same rank calculation used in Corollary 4.2. Induction over the finite tree gives the claim. $\square$

For the extreme-point statement, define the leaf contact set

$$T_{\ell}(f_{\ell}) = \left\{ t \in K_{\ell} : \int_0^t f_{\ell} = b_{\ell}(t) \right\}.$$

We need one elementary convex lemma.

Lemma 6.2 (one affine slice of a product). *Let C_i be convex sets, let ℓ_i be affine real functionals, and put*

$$C(c) = \left\{ (x_i) \in \prod_i C_i : \sum_i \ell_i(x_i) = c \right\}.$$

A point $x = (x_i)$ is extreme in $C(c)$ if and only if every x_i is extreme in its fixed-value fiber

$$\{y \in C_i : \ell_i(y) = \ell_i(x_i)\},$$

and at most one x_i is not extreme in the whole set C_i .

Proof. A nontrivial symmetric perturbation inside one fixed-value fiber immediately gives a perturbation in $C(c)$. If two coordinates admit whole-set perturbations but no fixed-fiber perturbations, both perturbations change their affine values. Reverse one and scale the two symmetric segments so that the changes cancel. Conversely, any nontrivial perturbation in $C(c)$ must, after fixed-fiber extremality is imposed, change the affine value in every perturbed coordinate. Since the changes sum to zero, at least two coordinates are perturbed. $\square$

Proposition 6.3 (leafwise extreme-point recursion). *For every node v of the tree, a feasible density is extreme in the whole subtree relaxation if and only if it is extreme in its fixed-total-mass fiber, if and only if every leaf below v satisfies*

$$f_{\ell} \in \{0, 1\} \quad \text{almost everywhere on } [0, 1] \setminus T_{\ell}(f_{\ell}).$$

Proof. At a leaf, the same perturbation and zero-set derivative argument as in Theorem 5.1 shows that whole-set and fixed-mass-fiber extremality coincide and are characterized by the displayed contact condition. At a direct-sum node, whole-set extremality is coordinatewise, while fixed-mass extremality is governed by Lemma 6.2. The induction hypothesis makes the lemma's extra alternative redundant. At a truncation node, an inactive capacity leaves the original convex set locally unchanged, while an active capacity replaces it by the corresponding fixed-mass fiber. The induction hypothesis again identifies the two extreme-point notions. $\square$

Finite laminar matroids are classically organized by direct sums, truncations, and related basic operations [8]. Propositions 6.1–6.3 should therefore be viewed as a measurable application of the preceding convex mechanism, not as a new structural classification of laminar matroids.

7 Scope and open boundary

Theorems 3.1 and 4.1 give exact infinite-dimensional counterparts of familiar finite lattice-path and positroid polytope descriptions. The finite shadow is not new: base polytopes of lattice-path matroids have path-inequality descriptions [2], and general positroid independence polytopes admit interval-type facet descriptions [3]. The new claim made here is limited to the atomless measurable setting, arbitrary closed K , the explicit countable purification and residual-capacity extension, and all real truncation levels. No claim of absolute global priority is made without external review.

Two natural questions remain.

1. Give a complete extreme-point criterion for the mass- c slice of $\mathcal{I}_K(a, b)$ when $c < R$. The equalities on active intervals give necessary perturbation conditions, but nonactive slacks may converge to zero, so the obvious finite-dimensional kernel argument is not sufficient.
2. Determine when finite laminar compositions as in Section 6 are genuinely outside the class of measurable lattice-path matroids. The classical finite truncation of $U_{1,2}^{\oplus 3}$ is not transversal and hence not a lattice-path matroid [9], but transferring this into a sharp atomless nonrepresentation theorem requires a separate argument.

These questions require new structure beyond the direct-sum and truncation identities used here.

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